ANITA MADANI

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EXPERIENCE

Machine Learning Research Intern (MLR)

Apple – Video Computer Vision (VCV) Team

May 2026 – Sep 2026 SunnyVale, CA

- Conducting research on video understanding and vision language models within the Video Computer Vision (VCV) team.
- Designing scalable deep learning architectures for multimodal representation learning and temporal reasoning.

Research intern

NEC Laboratories America INC - Media Analytics Lab

Feb 2026 - May 2026

- Architecting generative agent-based AI models using diffusion frameworks to create high-fidelity, reactive simulations for realistic scene synthesis and trajectory forecasting.
- Developing data-driven frameworks for modeling pedestrian dynamics and large-scale human behavior patterns to improve autonomous vehicle predictability.

Graduate Research Assistant

Johns Hopkins University, AIEM & VIU

2024 – Present Baltimore, MD

- Developed deep learning models for **image registration**, **visual localization**, and **3D reconstruction** under challenging real-world conditions.
- Researched **remote sensing** and **change detection** using multimodal representation learning to resolve temporal and seasonal variations.
- Built **diffusion** and **vision-language models** for large-scale scene understanding; collaborated on the multi-institutional WRIVA project.

SKILLS

Coding

Python (50k+ lines) | C (10k) | C++ (10k) | Bash (5k) | Java (5k) | MATLAB (3k) | Verilog (1k)

Expertise

Computer Vision | 3D Reconstruction | Visual Localization | Change Detection | Remote Sensing | Deep Learning | Diffusion/Generative Models | LLMs | Multi-Modal Learning | Algorithms & DS | Optimization | Linear Algebra | Probability | Signal & Image Processing

Tools

PyTorch | TensorFlow | Keras | scikit-learn | Hugging Face | ONNX | OpenCV | COLMAP | torchvision | TensorBoard | MLflow | Git | Docker | SLURM | Conda/Mamba | CMake | Linux | AWS | SQL

HONORS & AWARDS

- **Graduate Fellowship** — Whiting School of Engineering, Johns Hopkins University
- **Qualified for national Math and Informatics Olympiad** (2016)
- **Valedictorian** — Farzanegan 1 School (NODET, Middle & High School)

EDUCATION

Ph.D., Electrical & Computer Engineering

Johns Hopkins University

2023 – 05/2027 (expected) United States

MSE, Electrical & Computer Engineering

Johns Hopkins University

2023 - 2025 United States **GPA:** 3.92

B.Sc., Electrical Engineering

Sharif University of Technology

2019 – 2023 Iran **GPA:** 3.74

SELECT PROJECTS & RESEARCH

Scene Graph Memory for Video Understanding

Apple

2026

- Developed a **VLM-based framework** for incremental scene graph construction and persistent object tracking.
- Improved graph accuracy by **10.7 points** and long-horizon VideoQA by **15.3 points** using **SFT and GRPO**.

Multimodal Agents for Driving Scene Simulation

NEC Laboratories America

2026

- Developed **multimodal agents** for pedestrian simulation using VLM planning, physics-based motion, and diffusion models.
- Achieved **70.3% behavior alignment** (vs. **14.9%** baseline) through VLM-based action verification and refinement.

WRIVA

Johns Hopkins University / IARPA

2024 – 2026

- Developed **cross-view image registration and 3D reconstruction** methods robust to viewpoint shifts across ground, UAV, and satellite imagery.

PUBLICATIONS

- SceneGraphMem: Learning Persistent Scene Graph Memory for Long-Horizon Video Understanding, *Preprint*, **CVPR 2027**.
- PedEditor: Pedestrian Trajectory and Behavior Editing via Generative Models, *Preprint*, **CVPR 2027**.
- IceBerg Attention: Sublinear Long-Context Decoding With Mixture-Of-Gaussians, *Under Review*, **ICLR 2027**.
- DiffRegCD: Integrated Scene Alignment and Change Detection Using Diffusion Features. **WACV 2026**.
- Morphing Through Time: Diffusion-Based Bridging of Temporal Gaps for Robust Alignment in Change Detection. **WACV 2026**.
- *On The Application of Linear Attention in Multimodal Transformers*. **CVPRW 2026**.